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Review of CEMS Regulations 40 CFR Part 60 vs Part 75

Select Acronyms

- CEMS – Continuous Emission Monitoring System
- CFR – Code of Federal Regulations
- CGA – Cylinder Gas Audit
- RATA – Relative Accuracy Test Audit
- RA – Relative Accuracy
- PS 2 – Performance Specification for SO₂ & NO_x Analyzers
- PS 3 – Performance Specification for CO₂ & O₂ Analyzers

U.S. EPA CEMS Regulations

- **40 CFR Part 60 CEMS Regulations**
 - First CEMS Regulations
 - Applied to All New CEMS Sources
- **40 CFR Part 75 Regulations**
 - Applies to Electric Power Plants and Select Fuel Combustion Plants in Eastern USA.
 - QA/QC Requirements More Strict Than Part 60 CEMS Regulations.

CEMS

PERFORMANCE SPECIFICATIONS

- **Performance Specification 1 Opacity Monitors**
- **Performance Specification 2 for SO₂ & NO_x**
- **Performance Specification 3 for O₂ & CO₂**
- **Performance Specifications (PS) include:**
 - **CEMS Location Requirements**
 - **Relative Accuracy Test specifications and procedures**
 - **Calibration Drift Test specifications and procedures**
 - **Equations**

CEMS PERFORMANCE SPECIFICATIONS

- **Procedures for Evaluating the Acceptability of CEMS at the Time of or Soon After Installation**
- **Quality Assurance/Quality Control (QA/QC) Regulations Found In Appendix F of Part 60 and Appendix B of Part 75**

Part 60 vs 75 Potential Quality Assurance Differences

- 40 CFR Part 60, Appendix B and F
- 40 CFR Part 75, Appendix A and B
- Part 60 *Calibration Drift* vs Part 75 *Calibration Error Check*
- Part 60 *Cylinder Gas Audit (CGA)* vs Part 75 *Linearity Check*

Part 60 vs 75 Potential Quality Assurance Differences (cont'd)

- **Relative Accuracy Test Requirements**
- **Reference Test Method Differences**
- **Flue Gas Moisture Basis**

CEMS Span and Range

Part 60 vs Part 75

- **Span is a subset of Range (range is always $\geq$ Span)**
- **Part 60 Sources – Usually Specified Span by the Subpart**
- **Part 75, Appendix A, Section 2, gives Span Calculations**
- **EPA's Objective to set range such that majority of measurement reading are in 20% to 30% range**

Load Range for Gas Monitor Relative Accuracy Test Audit (RATA)

40 CFR 60

- Appendix B, Performance Specification 2, Section 5.3
- Unit operating at more than 50% of normal load

40 CFR 75

- Appendix B, Section 2.3.13(a)
- Unit normal load specified by Appendix A, Section 6.5.2.1

Part 60 vs Part 75

Gas Monitor Relative Accuracy (RA)

40 CFR 60

- PS 2 RA $\leq$ 20% or 10% to 20% of the Standard
- PS 3 RA $\leq$ 20% or 1.0% O₂ or CO₂
- Repeat every four quarters (Appendix F)

40 CFR 75

- RA $\leq$ 10%, ± 15 ppm or $\pm 1.0\%$ O₂ or CO₂
- RA $\leq$ 10%, or $\pm 1.0\%$ O₂ or CO₂
- Repeat 2 QA operating quarters + grace period

Part 75 RATA Incentive

- $RA \leq 7.5\%$ pollutant monitors
- Low emitter standard ± 0.020 lb/mmBtu
- Repeat every four quarters plus grace period

Part 60 Calibration Drift vs Part 75 Calibration Error

40 CFR 60

- Performance Specification (PS) 2.5% of Span for NO_x and SO₂ or 0.5% O₂ or CO₂
- Out-of-Control Period – first failed 4 x PS or fifth consecutive failed 2 x PS (Appendix F)

40 CFR 75

- Performance Specification (PS) 2.5% of Span for NO_x and SO₂ (≤ 5 ppm for spans < 200 ppm), or 0.5% O₂ or CO₂
- 3.0% for Flow Monitors
- Out-of-Control Period - first failed 2 x PS

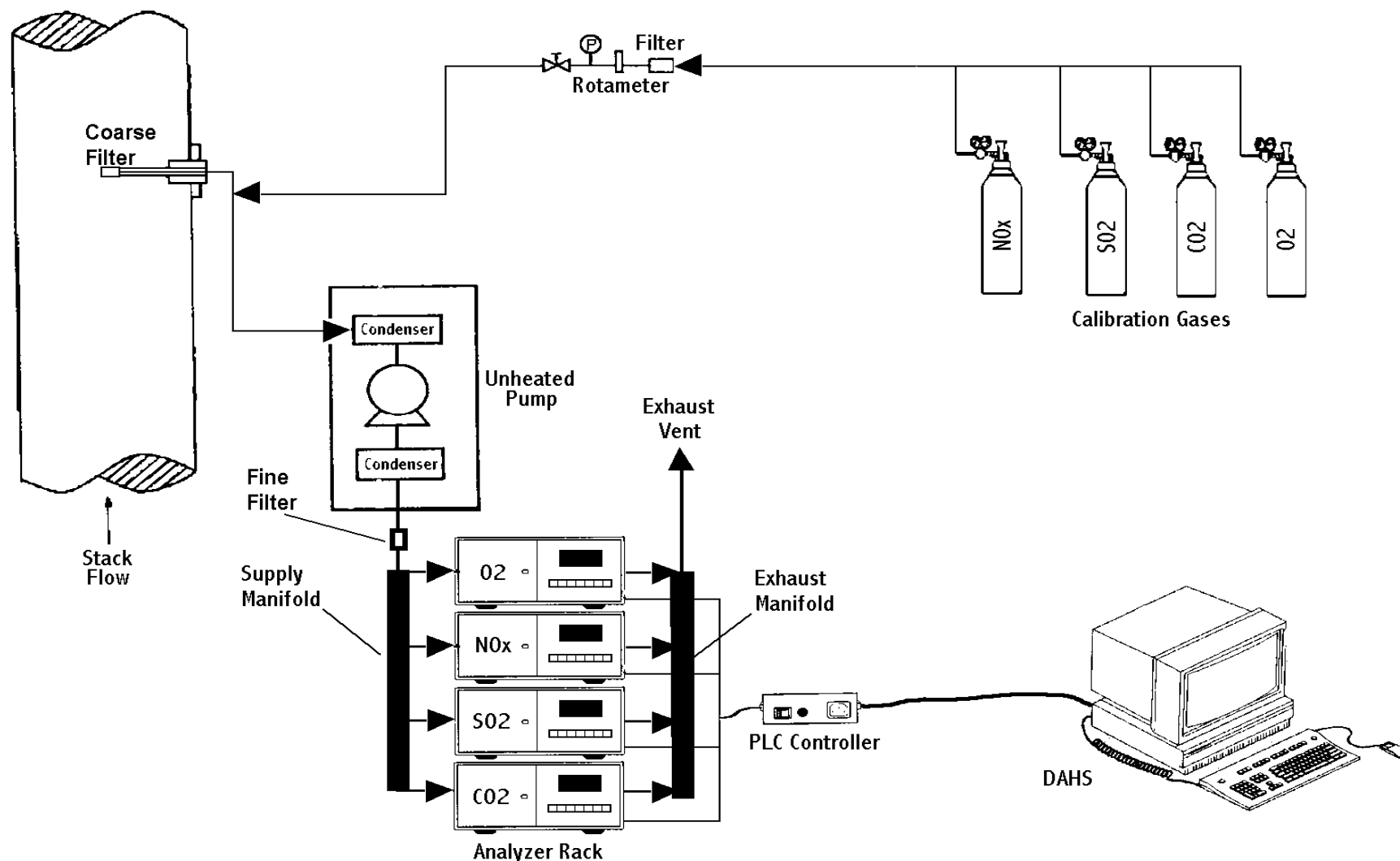
Part 75 - Calibration Error & Linearity Out-of-Control

- **When Daily Calibration Error or Linearity Check Is Found Out-Of-Control, The Emissions Data Is Invalid And Missing Data Substitution Must Be Used.**
- **Invalid Data Starts At The Time That Calibration Error or Linearity Showed Out-of-Control Status, And Remains Invalid Until Corrective Action Shows No Out-of-Control Status.**

Part 75 Calibration Error Test CEMS Design Requirements

- Design and equip each pollutant concentration and CO₂ or O₂ monitor with a calibration gas injection port that allows a check of the entire measurement system when calibration gases are introduced. For extractive and dilution type monitors, all monitoring components exposed to the sample gas, (e.g., sample lines, filters, scrubbers, conditioners, and as much of the probe as practicable) are included in the measurement system.
- “Very Important Requirement”

Recommended CEMS Calibration System



Part 75 Calibration Error Test CEMS Design Requirements

- For in-situ type monitors, the calibration must check against the injected gas for the performance of all active electronic and optical components (e.g. transmitter, receiver, analyzer).
- “Very Important Requirement”

Part 60 CGA vs Part 75 Linearity Check

40 CFR 60

- 2 calibration gases: 20-30% and 50-60% of monitor span
- Criteria $\leq \pm 15\%$ or ± 5 ppm
- 3 quarters per year – any quarter with RATA is exempt (Appendix F)

40 CFR 75

- 3 calibration gases: 20-30%, 50-60%, & 80-100% of monitor span
- Criteria $\leq \pm 5\%$ or ± 5 ppm
- Every QA operating quarter + grace period

Calibration Gases for Reference Methods

40 CFR 60

- Part 60, Appendix A, requires tester to use protocol gases or may elect alternative no. 2

40 CFR 75

- Part 75, tester must use gases in Section 5, Appendix A of Part 75

CEMS Location Specifications

(Same for Part 60 & 75)

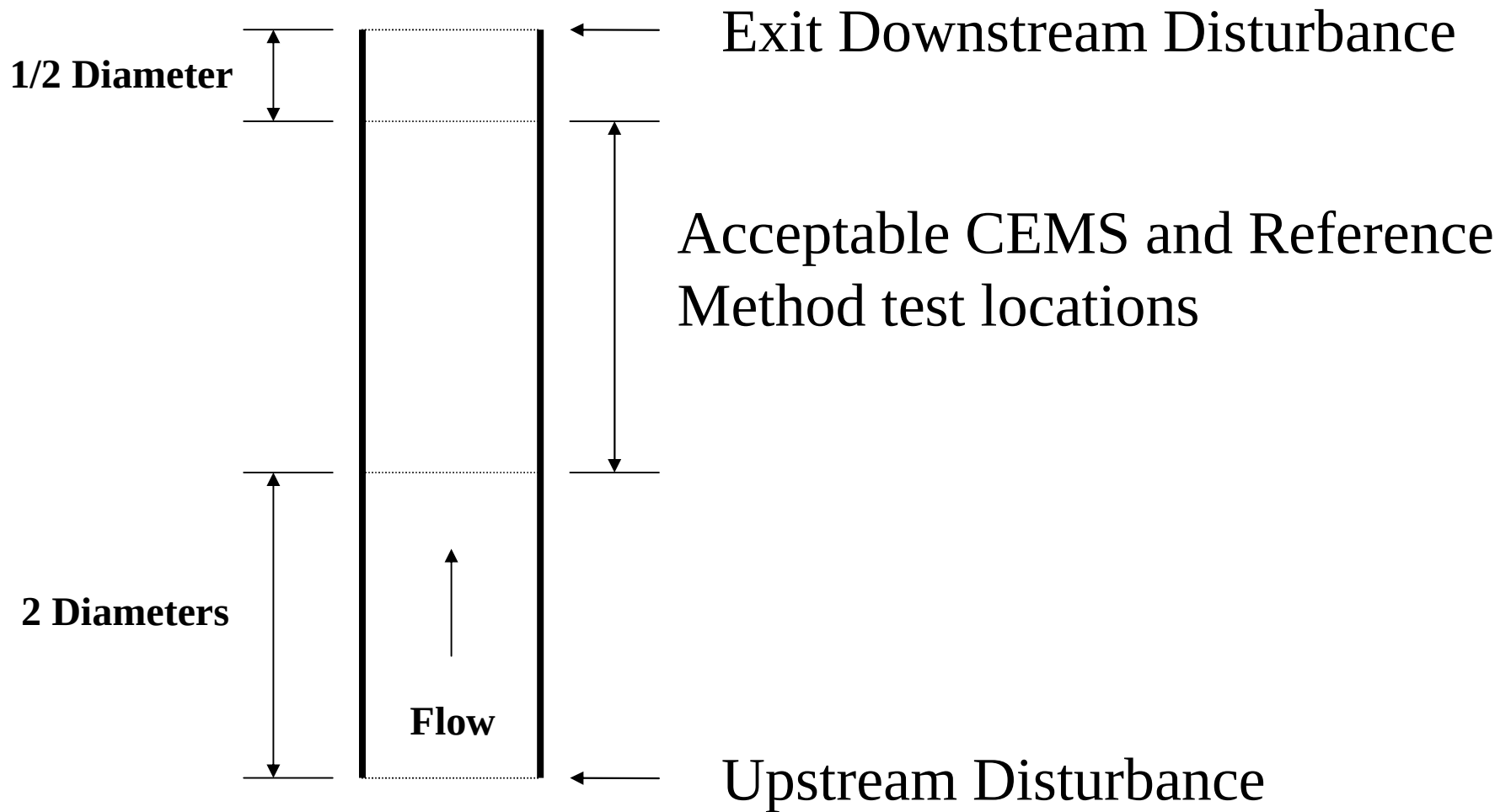
- CEMS must be accessible, representative, and capable of passing a Relative Accuracy Test
- At least 2 duct/stack diameters downstream of any “disturbance”
- At least 1/2 diameter upstream of any “disturbance”
- Any other location demonstrated to be acceptable

CEMS Location Specifications

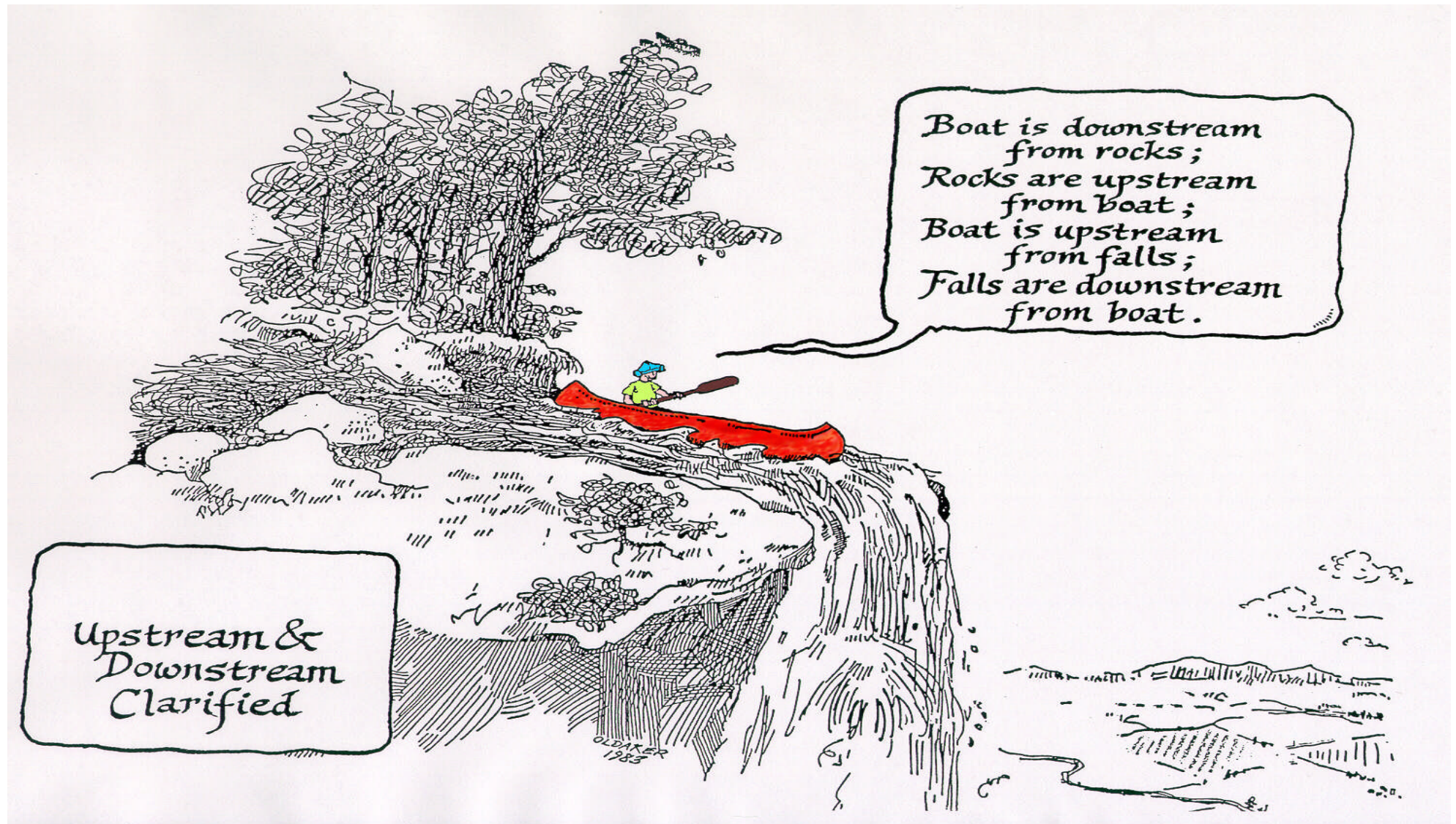
(Same for Part 60 & 75)

- “Disturbances” include:
 - A point of pollution generation
 - Control device inlet or outlet
 - Stack exit
- Reference Method (RM) test locations must be accessible and representative

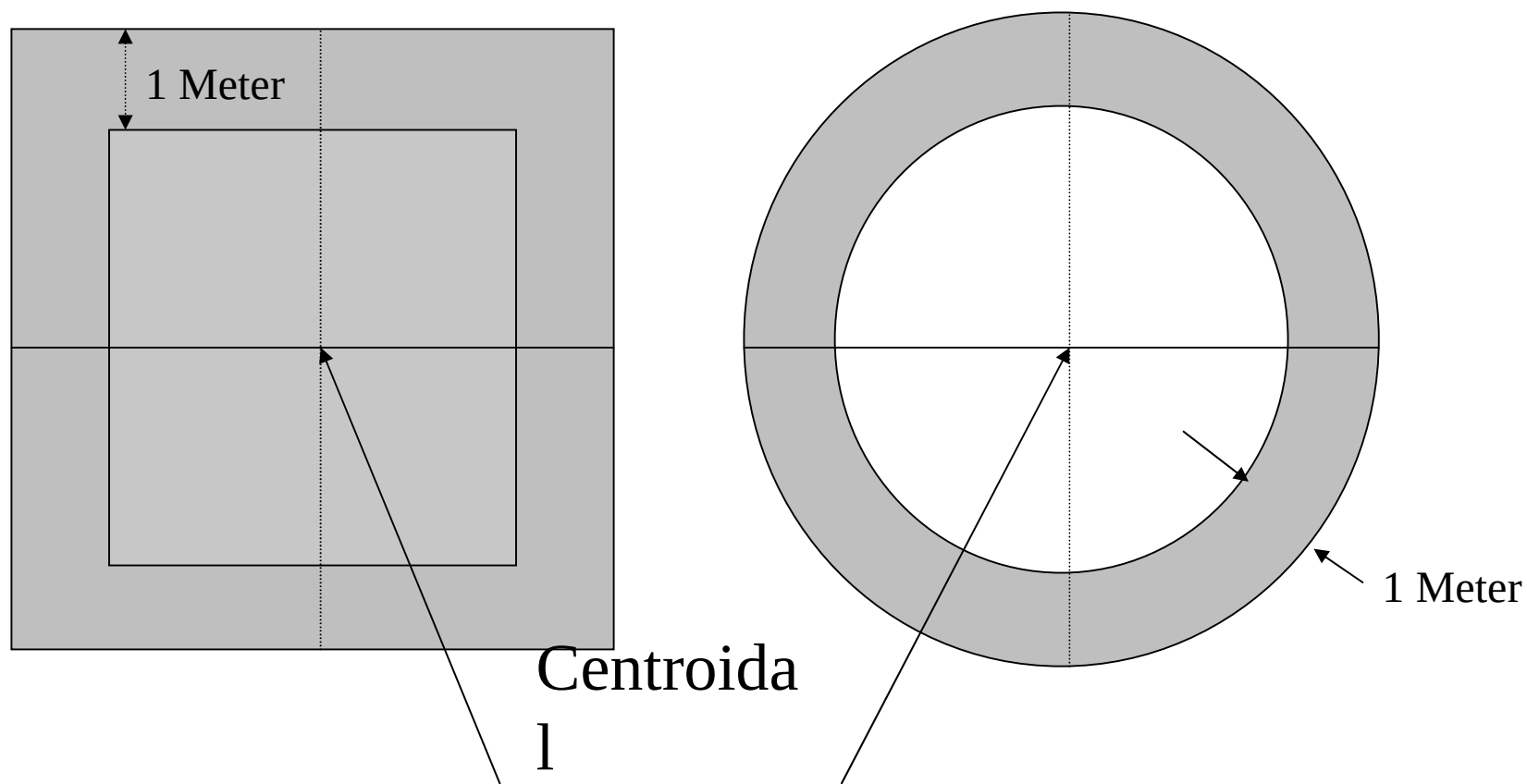
Recommended CEMS Locations



Upstream & Downstream Clarified



CEMS Sampling Probe Locations



The probe must be located at least 1 meter in from the duct/stack wall or at the centroid of the duct/stack.

Gas Monitor Relative Accuracy (RA) Part 60 vs Part 75

40 CFR 60

- PS 2 RA $\leq$ 20% or 10% to 20% of the Standard
- PS 3 RA $\leq$ 20% or 1.0% O₂ or CO₂
- Repeat every four quarters (Appendix F)

40 CFR 75

- RA $\leq$ 10%, $\pm$ 15 ppm or 1.0% O₂ or CO₂
- RA $\leq$ 10%, or $\pm$ 1.0% O₂ or CO₂
- Repeat 2 QA operating quarters + grace period

Part 75 RATA Incentive

- $RA \leq 7.5\%$ Pollutant Monitors
- Low Emitter Standard ± 0.020 lb/mmBtu
- Repeat every four quarters plus grace period

Reference Method Differences

40 CFR 60

- Test Methods specified in Subpart

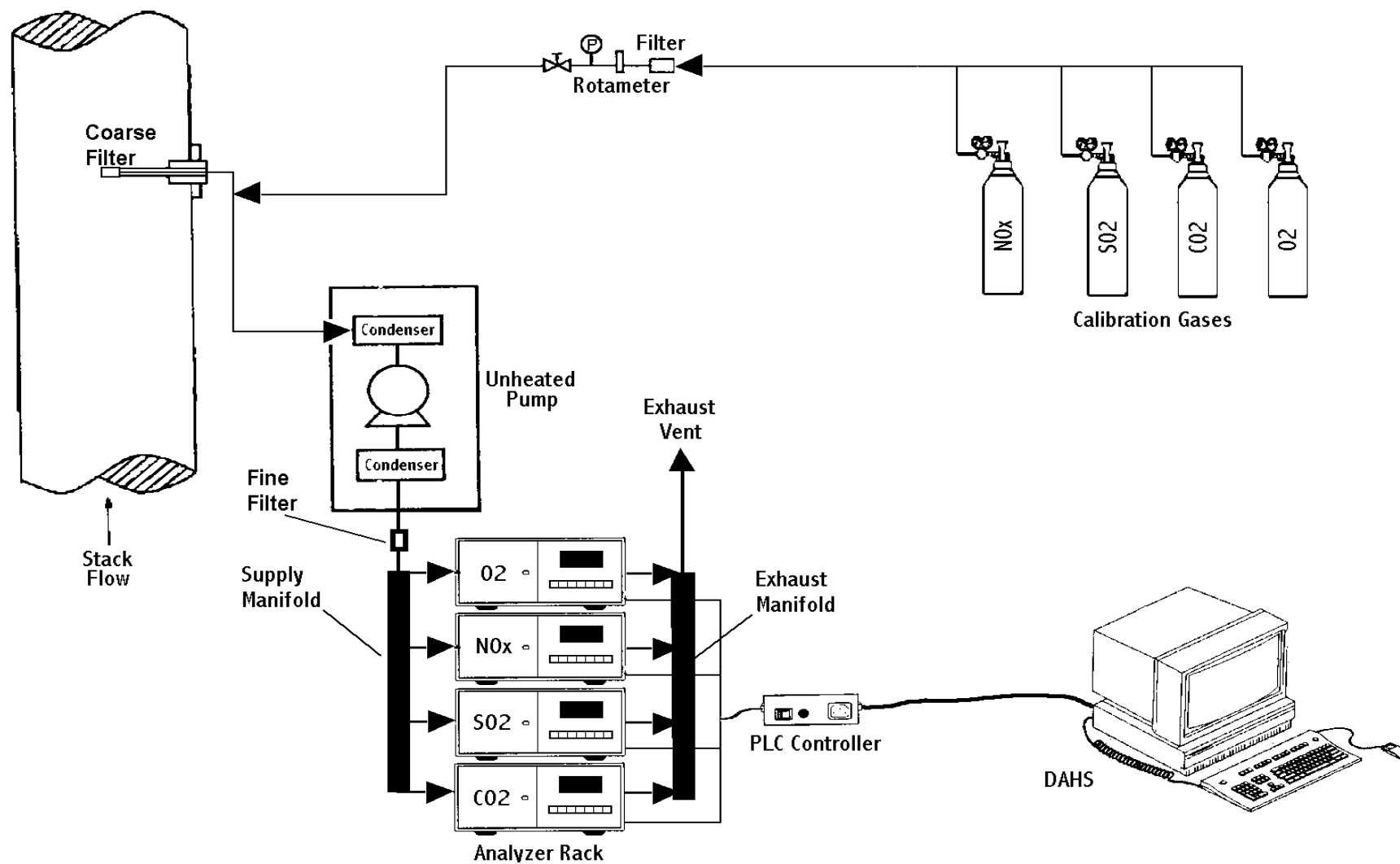
40 CFR 75

- Test Methods specified in §75.22. *Not all of the sections in the reference methods are adopted in Part 75 (See Appendix A, §6.5.10 for total NO_x*

EPA Reference Method Procedures

- Method 6C -- SO₂
- Method 7E -- NO_x
- Method 3A -- CO₂ and O₂
- Method 4 -- Moisture
- Other alternative Reference Methods exist but should not be used for Relative Accuracy Tests

Instrumental Test Methods



Relative Accuracy Test Probe Locations

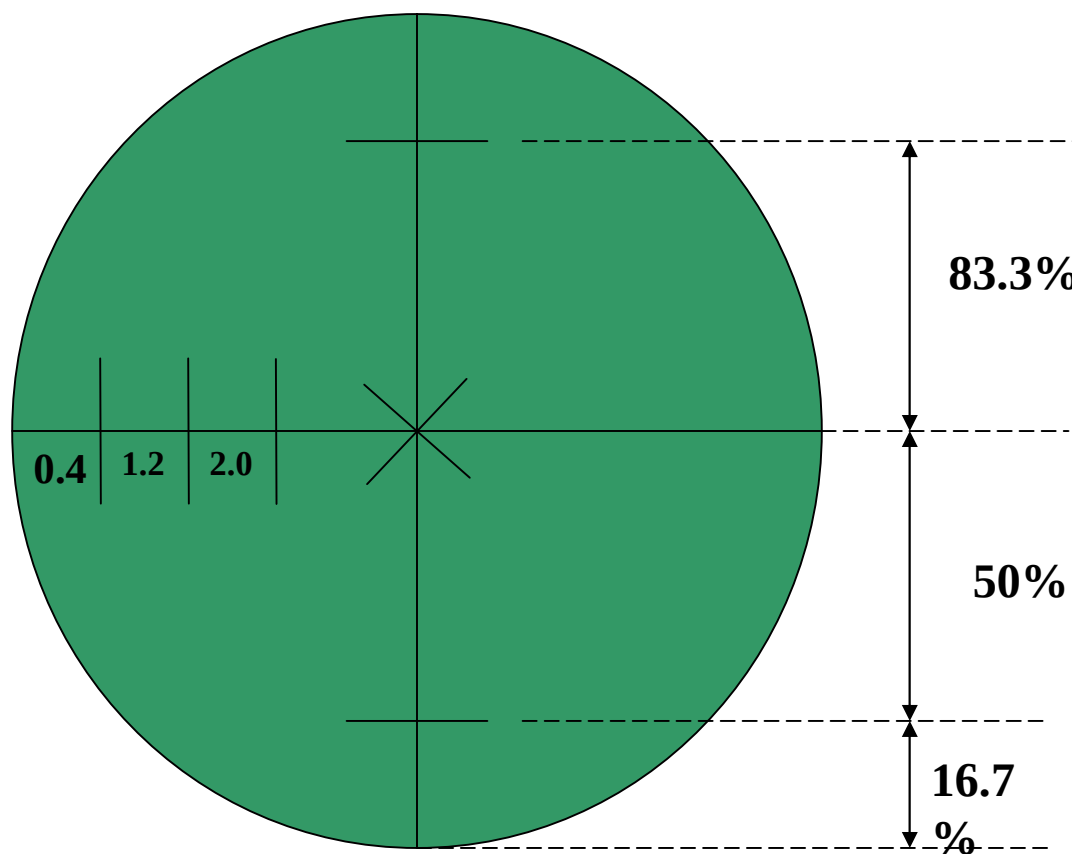
- Testing should be performed as close to the CEMS location as possible
- Three-point traverses are required
- Test crews are responsible for obtaining a representative sample

Reference Test Method Procedures

- Minimum test duration -- 21 minutes
- Sample Each Point for one third of the test run duration
- Grab samples must be collected at equally spaced time intervals throughout the test run
- Relative accuracy is based on 9 test runs
- Up to 3 test runs may be discarded -- but must be reported

Traverse Point Locations

Traverse points must be 0.4, 1.2 and 2.0 meters in from the stack or duct wall; or if downstream from a wet scrubber, 16.7, 50 and 83.3% of the stack or duct diameter.



Relative Accuracy Calculations

Where: $t_{0.975}$ = t-value obtained from the table in PS 2

n = the number of test runs

**Std. Dev. = the standard deviation of the differences
between the CEMS and the Reference
Method test run results**

$$\text{Confidence Coefficient (CC)} = \frac{t_{0.975} (\text{Std. Dev.})}{\sqrt{n}}$$

Relative Accuracy Calculations

$$\text{Relative Accuracy (RA)} = [(|M_d| + CC) / M_R]$$

**Where: M_d = Mean difference between the CEMS
and the Reference Method test results**

M_R = Mean Reference Method test value

7-Day Calibration Drift (Error) Test Procedures

- Unit must be operating at $> 50\%$ normal operating load
- Measure the calibration drift (CD) over 7 consecutive 24-hour periods (168 hours)
- If adjustments are made to the CEMS during this 7-day test, the magnitude of each adjustment must be recorded.

7-Day Calibration Error Test Procedures

Part 75

- Calibration error is determined at two concentrations for Gases-- 0 to 20% of Span and 80 to 100% of Span
- EPA Protocol 1 Calibration Gases Must Be Used
- Flow Monitor – 0 to 20% of Span and 50 to 70% of Span

- None of the CDs may exceed:
 - 2.5% of the Span Value for SO₂ and NO_x
 - 0.5% for O₂ and CO₂
 - 3.0% for Flow Monitors

Part 75 QA/QC Requirements

- **QA Plan must be prepared prior to initial certification**
- **Relative Accuracy Test Audits (RATAs)**
- **Calibration Error Tests**
- **For flow monitors, Interference Tests, and in some cases, Leak Checks**
- **Linearity Checks**

Quality Assurance & Quality Control

- **Quality Assurance & Quality Control (QA/QC)**
 - **Quality Assurance is "the system of activities to provide assurance that the Quality Control system is performing adequately," whereas Quality Control is "the system of activities to provide a quality product (e.g., environmental measurements)."**

- **Quality Assurance is often used to define "external" activities (i.e., functions performed on an occasional basis, such as QA audits, usually by someone not otherwise involved with routine CEMS operations).**

- **Quality Control is often used to characterize "internal" activities (e.g., functions conducted on a more frequent basis, such as periodic calibrations and routine maintenance procedures, usually performed by someone whose principal responsibilities include CEMS operation and maintenance).**

Quality Control & Quality Assurance (QA/QC)

- **Together, Quality Assurance and Quality Control QA/QC Form A Control Loop That Ensures CEMS Emissions Data Acceptability.**